



Research paper

SMEs and artificial intelligence (AI): Antecedents and consequences of AI-based B2B practices

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ABSTRACT

Development of small and medium enterprises (SMEs) is a key approach to achieving economic growth in the Middle East and successful adoption of technology is vital for SMEs' success and continuity. Artificial intelligence (AI) is part of a new generation of technologies that can facilitate competitive advantage but currently there is a lack of evidence regarding AI applications in relation to B2B SMEs in Middle East countries. Therefore, this study empirically examines antecedents to, and consequences of, successful acceptance of AI practices by B2B SMEs in Saudi Arabia. A conceptual model based on the technology-organisation-environment framework is developed which considers the impact of AI enablers and AI readiness on the acceptance of AI practices, and the impact of this on relational governance, performance, and SMEs' AI-based business customer interaction. The conceptual model was tested using structural equation modelling of survey data collected from B2B SMEs ($n = 392$). The results showed that, of the AI enablers, acceptance of AI practices was significantly influenced by both technology roadmapping and attitude but not professional expertise. Of the AI readiness variables, acceptance of AI practices was significantly influenced by infrastructure and awareness but not technicality. The acceptance of AI practices was found to significantly affect AI-enabled relational governance and performance, and SME's business customer AI-based interaction. This study provides a broader base for theoretical and practical understanding of issues related to AI practices in SMEs and the B2B sector in general.

1. Introduction

Small and medium enterprises (SMEs) account for the majority of businesses globally (World Bank, 2020) and they have “become the centrepiece of strategies to achieve inclusive growth in the Middle East, North Africa, Afghanistan and Pakistan (MENAP) region” (Lukonga, 2020, p.5). SMEs have been increasingly extending their activities within business markets in order to create more opportunities for growth and expansion (Mogaji, Soetan, & Kieu, 2020). B2B SME success largely depends on the extent to which there is an effective level of coordination and contact between SMEs and their business customers (Osterwalder & Pigneur, 2010; Rauch et al., 2018). However, the shape and nature of

B2B interactions seem to be more complex and intertwined than in the consumer market (Agnihotri, Dingus, Hu, & Krush, 2016), thus creating a need for sophisticated solutions.

Adoption of emerging technologies has contributed considerably to SMEs' ability to effectively overcome challenges, collaborate and interact with their business customers, and improve firm performance (Agnihotri et al., 2016; Brink, 2017; Dwivedi & Mustafee, 2010; Ghobakhloo & Ching, 2019; Sila & Dobni, 2012; Tan, Chong, Lin, & Eze, 2010). Artificial intelligence (AI) is a system's ability to interpret and learn from data (Aker et al., 2021; Alter, 2021; Haenlein & Kaplan, 2019), mimicking human intelligence (Rahman, Hossain, & Fattah, 2021), and it is part of a new generation of technologies that introduce

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